

2026. 9. 15

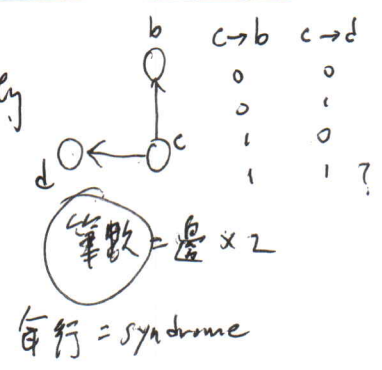
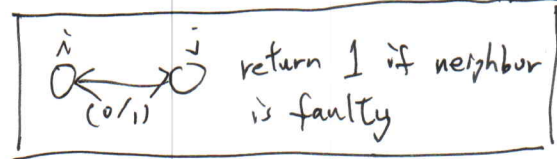
王大陸, 網路診斷與容錯

Motivation & Background

1. More high speed large scale multi processor / computer systems being deployed using interconnection networks of different topology
2. naming conventions
 - binary
 - 4-digit
 - hypercube
 - star graph
 - k-ary n-cube
 - bubble sort graph
3. Networks grow in size, more vulnerability & attacks
4. Needs methods to find multiple faults in system → System-Level Diagnosis
= mutually check all units in system and identify faults / bad nodes

Mutual Diagnosis

1. perform test on neighbor nodes and determine good/bad (0/1)
2. invalid result if tester node is bad (result can be floating)
3. through multiple test systemwide identify floating edge as faulty (all edges coming out of faulty node floating)
4. $\begin{cases} \text{good} \rightarrow \text{bad} = 1, \text{ certain} \\ \text{good} \rightarrow \text{good} = 0, \text{ certain} \\ \text{bad} \rightarrow \text{any} = \text{uncertain } x(0/1) \end{cases}$
5. 結果不重合 → 找壞點



PMC Model: Preparata - Metze - Chien

1967, On the connection assignment problem of diagnosable systems

1. 壞點不能多 → 結果發散
2. t-Diagnosability: 存在一數 t, 壞點數 > t 時方法失效 → 描述診斷度
3. t 超過時現象: 不同壞點時產生相同檢測結果
4. 結果表格巨大不實際 → 新方法描述診斷 →
5. 對壞點分布做合理假設可提高診斷度 (前面定義 t 對分布無限制)

$$|V| = 2t + 1$$
$$k(G) \geq t$$

$|V|$: total nodes
 $k(G)$: min degree of G

Conditional Diagnosability

1. 假設某結果不會出現 → 能排除很多錯誤結果
2. hypercube: $\begin{cases} 4n - 7 & \text{for } n \geq 5 \\ n & \text{other} \end{cases}$ conditional